MTH633 Group Theory

Question No : 11 of 41

$(Z, +)$ is ---------- of $(R, \cdot)$

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page 305

Answer (Please select your correct option)

- subgroup
- not subgroup
- trivial subgroup
- non-trivial subgroup

MTH633 Group Theory

Question No : 12 of 41

Every subgroup of a cyclic group is ---------- group.

page 385

Answer (Please select your correct option)

- Cyclic
- Non- cyclic
- Non-abelian
- Finite

MTH633 Group Theory

Question No : 13 of 41

S_3 is NOT the group D_3 of symmetries of an equilateral triangle.

Type text here

page 441

Answer (Please select your correct option)

- Yes
- No



Question No : 11 of 41

$\prod_{i=1}^{n} G_i$ is the ---------- notation for all the group operations for $G_1, \cdots, G_n$.

page 484

Answer (Please select your correct option)

- additive
- multiplicative
- subtractive
- division

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14

Question No : 15 of 41

$Z_2 \times Z_3$ is cyclic group with generator ----------

page 496

Answer (Please select your correct option)

- (1, 1)
- (0, 0)
- (1, 0)
- (0, 1)

MTH633 Group Theory

Question No: 10 (Marks: 1)

Let H be a subgroup of G and a is in G, then _____

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Answer (Please select your correct option)

- $|aH| = |bH|$ ⇦
- $|aH| \neq |bH|$
- $|aH| < |bH|$
- $|aH| > |bH|$

MTH633 Group Theory

Question No : 17 of 41

If N is _____ subgroup of a group G, the left cosets of N in G are the same as the right cosets of N in G.

page 652

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Answer (Please select your correct option)

- normal
- proper
- improper
- trivial

Let φ be a homomorphism function from group $(G,\bullet)$ to group $(G',\cdot)$ Then, $(Ker\,\varphi,\bullet)$ is _____ subgroup of $(G,\bullet)$

page 675

Answer (Please select your correct option)

- commutative
- factor
- simple
- normal

MTH633 Group Theory

Question No : 19 of 41

$Ker(h)$ is called the null space of A if it consists of all $v \in \mathbb{R}^n$ such that _____

page 700

Answer (Please select your correct option)

- $Av = a$
- $Av = b$
- $Av = c$
- $Av = 0$ ⇦

MTH633 Group Theory

Question No : 20 of 41

Let the complex numbers of magnitude 1 form a subgroup U of C^*, then the cosets of U are _____

page 718

Answer (Please select your correct option)

- Parallel lines
- circles with center at the origin. ⇦
- Non-parallel lines
- None of these

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Amir Vu
August 26, 12:11 PM

MTH633 Group Theory

Question No : 1 of 41

A_3 is a normal subgroup of S_3

page 919

Answer (Please select your correct option)

Yes

No

MTH633 Group Theory

Question No : 2 of 41

Marks: 1 (Budgeted Time: 1 Min)

Generally, for any sets A, B, and C, we can view a map * A x B→ C as defining a "multiplication," where any element a of A times any element b of B has as value some element c of C

page 946

Answer (Please select your correct option)

True

MTH633 Group Theory

Question No : 3 of 41

Sometimes a set X is used to study G via a group action of G on X

page 953

Answer (Please select your correct option)

- Yes
- No



MTH033 Group Theory

Question No : 4 of 41

Let G be a group acting on a set X, and let $x \in X$. Then the set $Gx = \{ax \mid a \in G\}$

is called the orbit of x in G

page 976

Answer (Please select your correct option)

- True
- False

Question No : 5 of 41

If H and N are subgroups of a group G, then the their join H V N is the smallest subgroup of G containing both H and N

page 1004

Answer (Please select your correct option)

- True
- False

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Question No : 6 of 41

For a prime number p, every group G of order p^2 is _____.

page 1091

Answer (Please select your correct option)

- abelian
- non-abelian
- infinite
- none of these

MTH633 Group Theory

Question No : 7 of 41

Each orbit in G under conjugation by G is a _____ in G.

page 1082

Answer (Please select your correct option)

- conjugate class
- subgroup
- trivial group
- none of these

MTH633 Group Theory

Question No : 8 of 11

A _____ contained in no larger p-subgroup.

page 1064

Answer (Please select your correct option)

- p - subgroup
- cyclic subgroups
- improper subgroups
- trivial subgroups

MTH633 Group Theory

Question No : 9 of 41

A non-abelian group G may have _____ of some order d dividing $|G|$.

page 1035

Answer (Please select your correct option)

- abelian subgroup
- subset
- no subgroup
- none of these

MTH633 Group Theory

Question No : 10 of 41

Let H be a subgroup of G and let N be a normal subgroup of G Then $(HN) / N =$ _____

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page 1009

Answer (Please select your correct option)

- $H / (H \cup N)$
- $H / (H \cap N)$
- $H \cup (H \cap N)$
- $H \cap (H \cap N)$